Light

# GCSE Computer Science — Network Protocols (Set 3)

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

The internet relies on a set of protocols that govern how data is transmitted between computers.
What is this protocol suite called?

- A. TCP/IP
- B. HTML/CSS
- C. USB/HDMI
- D. RAM/ROM

Q2 (1 mark)

When browsing the internet, a protocol transfers web pages from servers to browsers.
Which protocol performs this function?

- A. HTTP
- B. FTP
- C. SMTP
- D. POP3

Q3 (1 mark)

An office worker composes an email and clicks the send button.
Which protocol transmits this email to the mail server?

- A. IMAP
- B. HTTP
- C. SMTP
- D. FTP

Q4 (1 mark)

A manager reads emails on their office computer, then later checks the same emails on their phone while commuting.
Which protocol allows emails to remain on the server for access from multiple devices?

- A. FTP
- B. HTTP
- C. SMTP
- D. IMAP

Q5 (1 mark)

A photographer needs to send a large collection of high-resolution images to a printing company's server.
Which protocol is designed for transferring files like this?

- ○ **A.** SMTP
- ○ **B.** FTP
- ○ **C.** POP3
- ○ **D.** HTTP

Q6 (1 mark)

A bank's website shows a padlock symbol and 'https://' in the browser address bar.
What does this indicate about the connection?

- ○ **A.** The website loads faster than normal
- ○ **B.** The connection is encrypted for security
- ○ **C.** The website contains no advertisements
- ○ **D.** The website is owned by the government

Q7 (1 mark)

An email protocol downloads messages to a user's computer and removes them from the server.
Which protocol behaves this way?

- ○ **A.** POP3

- B. SMTP
- C. HTTP
- D. IMAP

Q8 (1 mark)

A network technician connects computers using cables and network switches.
Which technology standard governs this wired network connection?

- A. Wi-Fi
- B. Bluetooth
- C. Ethernet
- D. HTTP

Q9 (1 mark)

A tablet connects to the internet through a wireless router without any physical cables.
Which technology enables this connection?

- A. Ethernet
- B. FTP
- C. SMTP
- D. Wi-Fi

Q10 (1 mark)

The TCP/IP model organises network functions into separate layers. Which layer is responsible for the physical transmission of data on a local network?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q11 (2 marks)

A web browser, email client, and file transfer program all communicate over a network.
At which TCP/IP layer do these applications operate?

- ○ **A.** Application layer
- ○ **B.** Link layer
- ○ **C.** Internet layer
- ○ **D.** Transport layer

Q12 (2 marks)

A packet needs to travel from London to New York, passing through multiple networks.
Which TCP/IP layer handles the addressing and routing for this journey?

- ○ **A.** Internet layer

- B. Application layer
- C. Transport layer
- D. Link layer

Q13 (2 marks)

A layer of the TCP/IP model provides reliable delivery by using acknowledgments and retransmitting lost data.
Which layer provides this functionality?

- A. Application layer
- B. Internet layer
- C. Link layer
- D. Transport layer

Q14 (2 marks)

A user enters their credit card details on a website using HTTPS.
How does HTTPS protect this information?

- A. HTTPS stores credit card numbers locally so they don't travel over the network
- B. HTTPS encrypts all data transmitted between the browser and server, preventing interception
- C. HTTPS blocks hackers from accessing the website entirely
- D. HTTPS makes the website load faster, reducing the time data is vulnerable

Q15 (2 marks)

A company's email system uses SMTP, POP3, and IMAP.
Which statement correctly describes what SMTP does?

- ○ **A.** SMTP downloads emails to mobile devices
- ○ **B.** SMTP synchronises emails across multiple devices
- ○ **C.** SMTP transmits outgoing emails from clients to servers and between servers
- ○ **D.** SMTP stores emails on the server for later retrieval

Q16 (2 marks)

At different TCP/IP layers, data is referred to by different names.
Which correctly matches the layer to its data unit name?

- ○ **A.** Transport layer: frames
- ○ **B.** Internet layer: segments
- ○ **C.** Link layer: packets
- ○ **D.** Transport layer: segments

Q17 (2 marks)

An office is deciding between Ethernet and Wi-Fi for connecting desktop computers.
Which statement correctly compares these technologies?

- ○ **A.** Ethernet requires physical cables but typically offers more stable and consistent performance

- ○ **B.** Wi-Fi provides faster speeds than Ethernet in all situations
- ○ **C.** Ethernet cannot be used for internet access, only for local networks
- ○ **D.** Wi-Fi is more secure because wireless signals cannot be intercepted

Q18 (2 marks)

A user's laptop crashes and all local data is lost.
If they were using POP3 for email, what is the likely consequence?

- ○ **A.** All emails are safely stored on the server and can be recovered
- ○ **B.** Emails downloaded to the laptop are lost because POP3 removes them from the server after download
- ○ **C.** The email account is automatically transferred to a new device
- ○ **D.** POP3 automatically backs up emails to cloud storage

Q19 (2 marks)

A student asks why networks need standardised protocols.
Which answer best explains the purpose of protocols?

- ○ **A.** Protocols make computers run faster by optimising processor usage
- ○ **B.** Protocols provide agreed rules that allow different devices and systems to communicate with each other
- ○ **C.** Protocols reduce the cost of network equipment by using simpler components
- ○ **D.** Protocols limit the number of devices that can connect to a

network

Q20 (2 marks)

A network technician needs to understand addressing at different TCP/IP layers.
Which statement is correct?

- ○ **A.** MAC addresses are used at the Internet layer for routing between networks
- ○ **B.** IP addresses are used at the Link layer for local delivery
- ○ **C.** Port numbers are used at the Transport layer to identify applications
- ○ **D.** All layers use the same type of address for simplicity

Q21 (2 marks)

A security analyst discovers that a company is using standard FTP for transferring confidential documents.
Why might this be a security concern?

- ○ **A.** FTP can only transfer files under 1MB in size
- ○ **B.** FTP transmits files and login credentials without encryption, allowing interception
- ○ **C.** FTP files become corrupted during transfer
- ○ **D.** FTP only works on Windows operating systems

Q22 (2 marks)

An online multiplayer game needs to transmit player positions to all participants in real time.
Which Transport layer protocol would be most appropriate?

- ○ **A.** TCP, because all data must arrive in the correct order
- ○ **B.** UDP, because speed is more important than guaranteeing every packet arrives
- ○ **C.** HTTP, because games use web technologies
- ○ **D.** SMTP, because it handles real-time messaging

Q23 (2 marks)

Data is being sent from a web browser to a web server.
In which order does data pass through the TCP/IP layers on the sending computer?

- ○ **A.** Link, Internet, Transport, Application
- ○ **B.** Transport, Internet, Application, Link
- ○ **C.** Application, Transport, Internet, Link
- ○ **D.** Internet, Link, Application, Transport

Q24 (2 marks)

A company needs to choose the correct protocol for different tasks.
Which matches all protocols correctly to their functions?

- ○ **A.** FTP: sending emails, SMTP: web browsing, HTTP: file transfer
- ○ **B.** HTTP: web browsing, SMTP: sending emails, FTP: file transfer
- ○ **C.** SMTP: file transfer, HTTP: sending emails, FTP: web browsing

- ○ **D.** HTTP: file transfer, FTP: sending emails, SMTP: web browsing

Q25 (2 marks)

As data travels down through TCP/IP layers, each layer wraps the data from the layer above with its own header.
What is this process called?

- ○ **A.** Compression
- ○ **B.** Encryption
- ○ **C.** Encapsulation
- ○ **D.** Fragmentation

Q26 (3 marks)

A user sends a file from their computer to a cloud storage server on a different network.
Explain how each TCP/IP layer contributes to this transmission.

- ○ **A.** Application: handles physical cables; Transport: assigns MAC addresses; Internet: formats the file; Link: uses FTP protocol
- ○ **B.** Application: uses FTP to manage file transfer; Transport: uses TCP for reliable delivery with segments; Internet: uses IP addresses for routing across networks; Link: creates frames with MAC addresses for local transmission
- ○ **C.** Application: routes packets between networks; Transport: handles physical signals; Internet: uses port numbers; Link: formats the file for transfer
- ○ **D.** Application: assigns IP addresses; Transport: creates frames; Internet: uses FTP; Link: handles TCP reliability

Q27 (3 marks)

A healthcare organisation currently uses:
• HTTP for patient portal login
• Standard FTP for transferring medical records
• Unencrypted email (SMTP without TLS) for appointment confirmations
What security risks does this create for patient data?

- ○ **A.** No significant risks; these are standard protocols used by many organisations
- ○ **B.** The only risk is slower performance compared to modern protocols
- ○ **C.** All three practices transmit sensitive patient data in plain text: HTTP exposes login credentials, FTP exposes medical records, and unencrypted email exposes personal details—all can be intercepted and read
- ○ **D.** The risk is limited to internal network users only; external attackers cannot exploit these protocols

Q28 (3 marks)

An IT administrator configures email for a new employee who will work from home, the office, and while travelling. The employee needs to:
• Access all emails from any location
• See consistent read/unread status across devices
• Keep emails organised in folders visible on all devices
Which email configuration meets these requirements?

- ○ **A.** POP3 for incoming email, with emails stored locally on each device
- ○ **B.** IMAP for incoming email, keeping emails on the server with synchronised status and folders across all devices
- ○ **C.** SMTP for both incoming and outgoing email to simplify configuration

○ **D.** HTTP for email access since it works in web browsers

Q29 (3 marks)

A network technician is diagnosing a problem. They discover that:
• The computer can access websites by IP address (e.g., typing 142.250.80.46 loads Google)
• The computer cannot access websites by name (e.g., typing www.google.com fails)
• The computer can ping local devices and the router successfully
At which TCP/IP layer is the problem occurring?

○ **A.** Link layer—the network cable must be faulty

○ **B.** Internet layer—IP routing is not functioning

○ **C.** Transport layer—TCP connections are being blocked by a firewall

○ **D.** Application layer—DNS name resolution is failing

Q30 (3 marks)

A development team must select appropriate Transport layer protocols for three features:
• Feature A: Secure file synchronisation where all data must arrive correctly
• Feature B: Real-time voice communication during video calls
• Feature C: Downloading software updates that must be verified
Which protocol selections are correct?

○ **A.** All three should use TCP because reliability is always the priority

○ **B.** All three should use UDP because speed is always more important

○ **C.** Features A and C need TCP for reliability; Feature B should use

UDP for low-latency voice

○ **D.** Feature A needs UDP for fast sync; Features B and C need TCP for quality

Total: **55** marks • Time: **30 min**

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